

Marketing Funnel & Conversion Analysis Report

Dataset: User Behavior Data

Analysis Tool: Python (Jupyter Notebook)

1. Introduction

The purpose of this analysis is to evaluate user engagement patterns within the platform and understand how users progress through the engagement funnel. By identifying drop-off points and measuring conversion rates, the business can implement targeted strategies to increase engagement, retention, and ultimately revenue.

The dataset includes anonymized user behavior information with 11 columns, including:

- **User ID** — unique identifier for each user
- **Device Model and Operating System** — device information
- **App Usage Time (min/day) and Screen On Time (hours/day)** — activity metrics
- **Battery Drain (mAh/day) and Data Usage (MB/day)** — device usage metrics
- **Number of Apps Installed** — user app behavior
- **Age and Gender** — demographic information
- **User Behavior Class / Stage** — the key engagement metric used to define funnel stages

The dataset contains **700 unique users**.

2. Funnel Definition

For the purposes of this analysis, the user behavior funnel is defined using the Stage column (numeric representation of engagement). The stages are interpreted as:

Stage Description

- 1 Low — minimal engagement (passive users)
- 2 Low-Medium — slightly more active
- 3 Medium — moderate engagement
- 4 Medium-High — higher engagement
- 5 High — very active users, most valuable

The funnel is hierarchical, moving from Stage 1 → Stage 5. The analysis focuses on the number of users at each stage, conversion rates, and drop-off points.

3. Data Overview

The distribution of users across stages is as follows:

Stage	Users
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1	136
2	146
3	143
4	139
5	136

Observation:

- The number of users fluctuates slightly across stages, with the largest group in Stage 2 and smallest in Stages 1 and 5.
- This suggests that while some users progress through the funnel, there are drop-offs at intermediate stages.

4. Funnel Analysis

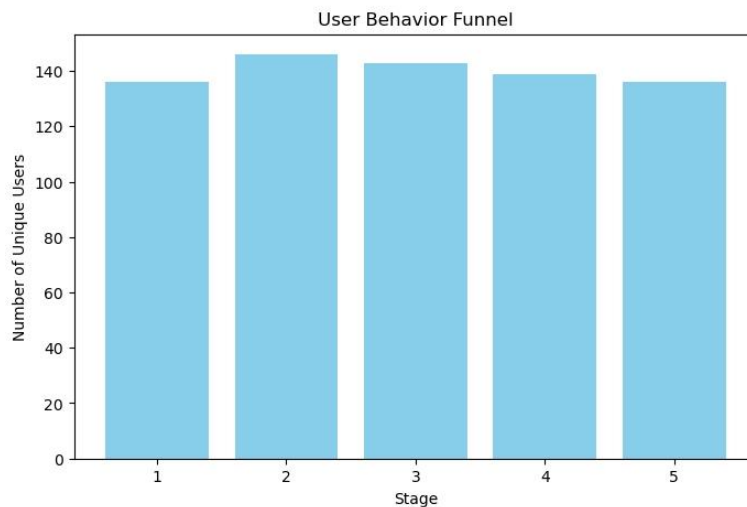
4.1 Conversion Rates & Drop-offs

Stage Transition	Conversion Rate (%)	Drop-off Users
Stage 4 → 3	2.88	-4
Stage 3 → 2	2.10	-3
Stage 2 → 5	-6.85	10
Stage 5 → 1	0.00	0

Insights:

- Conversion is inconsistent, with small increases between some stages and a notable drop-off moving into Stage 5.
- The largest drop-off occurs between Stage 2 → 5, suggesting user engagement strategies should focus on retaining mid-level users before they regress.
- Stage 1 users remain constant, indicating the base of low engagement users is stable.

5. Funnel Visualization



Interpretation:

- User numbers peak at Stage 2 and then decline toward Stage 5.
- Despite minor increases in early transitions, there is a significant drop at Stage 2 → 5, marking it as a critical point for engagement improvement.

6. Detailed Insights

1. Low Engagement Users (Stage 1)

- Represent a stable group of passive users.
- Action: Introduce onboarding tutorials, notifications, or gamification to encourage progression.

2. Medium Engagement Users (Stages 2–3)

- Largest user group; progression is inconsistent.
- Action: Personalized recommendations, prompts, or rewards to guide these users toward higher engagement.

3. Medium-High Engagement Users (Stage 4)

- Users here are moderately active but prone to regression.
- Action: Targeted incentives, early feedback, or premium features can encourage progression to Stage 5.

4. High Engagement Users (Stage 5)

- Smallest segment but highly valuable.
- Action: Offer loyalty programs, referral incentives, or exclusive features to maintain engagement.

5. Demographic and Device Insights

- Examine Age, Gender, Device Model, and OS to target specific groups with tailored strategies.
- Example: Users with low app usage on certain devices might benefit from app performance improvements.

7. Recommendations

1. Strengthen Mid-Funnel Retention

- Focus on users in Stages 2–4 to reduce drop-offs before reaching Stage 5.

2. Targeted Engagement for Low Users

- Onboarding guides, interactive tutorials, or initial incentives to convert Stage 1 → 2.

3. Nurture High Engagement Users

- Loyalty programs, premium offers, or referral incentives to maintain Stage 5 users.

4. Data-Driven Monitoring

- Continuous funnel tracking and A/B testing to evaluate engagement strategies and optimize interventions.

8. Conclusion

The analysis highlights that mid-funnel engagement is the key area of concern. While early-stage users are relatively stable, the critical drop-off occurs as users attempt to progress toward high engagement. Focusing on strategies that retain and motivate Stage 2–4 users will likely have the greatest impact on overall engagement and retention.